Two **completely different recovery scenarios**.

Scenario A: You accidentally run

docker compose down

without `-v`

What gets removed?

✅ Containers

❌ Volumes stay

❌ PostgreSQL data stays

❌ HDFS data stays

Recovery

Simply:

docker compose up -d

Then verify:

docker ps

Expected:

crime-intelligence-postgres
crime-hadoop-namenode
crime-hadoop-datanode

Check PostgreSQL

docker exec -it crime-intelligence-postgres \
psql -U anushka_admin -d crimedb

SELECT COUNT(*) FROM person;

Check Hadoop

docker exec -it crime-hadoop-namenode \
hdfs dfsadmin -report

Should still show:

Live datanodes (1)

This recovery takes about 30 seconds.

Scenario B: You accidentally run

docker compose down -v

This is the dangerous one.

What gets removed?

Containers:

❌ removed

Volumes:

❌ removed

PostgreSQL data:

❌ deleted

HDFS metadata:

❌ deleted

HDFS blocks:

❌ deleted

What survives?

Only files on your host machine:

Data_Generator/output_data/

Your CSVs survive.

Your source code survives.

Your schema survives.

Recovery Procedure

Step 1

Start infrastructure:

docker compose up -d

Step 2

Wait until healthy:

docker ps

Expected:

crime-intelligence-postgres

crime-hadoop-namenode
crime-hadoop-datanode

all healthy.

Step 3

Verify HDFS

docker exec -it crime-hadoop-namenode \
hdfs dfsadmin -report

Expected:

Live datanodes (1)

Step 4

Verify schema

docker exec -it crime-intelligence-postgres \
psql -U anushka_admin -d crimedb

\dt

Should show all 19 tables.

Since schema is mounted through:

docker-entrypoint-initdb.d

it will be recreated automatically.

Step 5

Reload PostgreSQL

Run:

python Data_Generator/load_data.py

Step 6

Verify counts

SELECT COUNT(*) FROM person;
SELECT COUNT(*) FROM case_details;
SELECT COUNT(*) FROM evidence;

Step 7

Recreate HDFS directories

docker exec -it crime-hadoop-namenode \
hdfs dfs -mkdir -p /crime_data/bronze/raw_csvs

Step 8

Copy CSVs back

docker cp Data_Generator/output_data/. \
crime-hadoop-namenode:/tmp/raw_csvs/

Step 9

Reload HDFS

docker exec -it crime-hadoop-namenode \
bash -c "hdfs dfs -put /tmp/raw_csvs/* /crime_data/bronze/raw_csvs/"

Step 10

Verify

```
docker exec -it crime-hadoop-namenode \
hdfs dfs -ls /crime_data/bronze/raw_csvs
```

Scenario C: Worst Case

If you run:

```
docker system prune -a --volumes
```

or

```
docker volume rm ...
```

the recovery procedure is exactly the same as Scenario B.

The reason is that your **CSV files are the source of truth**.

Everything else can be rebuilt from them.